

SM-8: Quark Generation Structure from 600-Cell Distance Shells

600-Cell Standard Model Emergence Series

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Abstract

The 600-cell admits exactly four bonded shells—the tetrahedron ($V=4$), icosahedron ($V=12$), dodecahedron ($V=20$), and icosidodecahedron ($V=30$)—separated by a structurally significant gap at Shell 3. These shells correspond one-to-one with the CPP cage types for quark generations. A zero-free-parameter mass formula

$$M_q = m_e \frac{z}{\varphi} V^{7/3} \quad (q = s, c, b); \quad M_t = m_e \frac{z}{\varphi} V_t^{7/3} \times z \cdot C_F$$

predicts all four quark masses to RMS 2.1% across four orders of magnitude, using only the electron mass, 600-cell geometry ($z = 12, \varphi$), and SU(3) colour algebra ($C_F = 4/3$). The post-gap multiplier is corrected from $z=12$ (v3.x) to $z \times C_F=16$ (v4.0) based on the angular-weighted pair model analysis. The distance shells exhibit palindrome symmetry, and antipodal identification in the tessellated lattice limits the Standard Model to exactly three generations.

Keywords: quark generations, 600-cell polytope, cage hierarchy, distance shells, Koide ratio, mass hierarchy, discrete spacetime, top quark, confinement, SU(3) Casimir

Plain Language Summary: The four types of “cages” that Conscious Point Physics uses to explain quark masses are built into the geometry of the 600-cell lattice. A formula using only the electron mass and lattice constants predicts all four heavy quark masses with no adjustable parameters.

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1 Introduction

The 600-cell is the unique four-dimensional regular polytope whose combinatorial structure supports both edge-based and face-based mode sectors relevant to the CPP framework. In earlier papers of this series (Abshier et al., 2026a,b), these sectors were shown to reproduce the charged-lepton and heavy-quark mass spectra through spectral traces, cage-level self-energy shifts, and isotropy theorems. The present paper examines a different geometric feature of the 600-cell: the **distance-shell structure** around a chosen vertex.

We show that the 600-cell contains **exactly four bonded shells**—the tetrahedron, icosahedron, dodecahedron, and icosidodecahedron—separated by a structurally significant gap at Shell 3, which contains no edges. These four shells correspond one-to-one with the four CPP cage types previously used to model quark generations. This correspondence is not imposed; it is a **forced combinatorial property** of the 600-cell.

Version 4.0 of this paper introduces a zero-free-parameter mass formula derived from the angular-weighted pair model (SM-9 v2.0), correcting the post-gap multiplier from $z = 12$ to $z \times C_F = 16$ and deriving both the scaling exponent ($\alpha = 7/3$) and the energy-scale prefactor ($M_0 = m_e z / \varphi$).

2 Physical Axioms for Cage-Scale Gauge Structure

Axiom A (Edge Abelianity). An edge-based interaction is modeled as transport along one-dimensional edge chains. Composition of transports along different edge paths is commutative: $U(\gamma_1)U(\gamma_2) = U(\gamma_2)U(\gamma_1)$. This defines an effective Abelian gauge structure at the cage scale.

Axiom B (Face Non-Abelianity). A face-based interaction is modeled as transport around closed triangular loops. The holonomy around two loops generally fails to commute: $U(\gamma_1)U(\gamma_2) \neq U(\gamma_2)U(\gamma_1)$. This path dependence motivates the identification of face modes with the colour sector.

3 The 600-Cell Distance Shells

Table 1: Distance shells of the 600-cell around any vertex.

Shell	d	$ \text{shell} $	Edges	z_{shell}	Structure
1	$1/\varphi \approx 0.618$	12	30	5	Icosahedron
2	1.000	20	30	3	Dodecahedron
3	≈ 1.176	12	0	0	(gap)
4	$\sqrt{2} \approx 1.414$	30	60	4	Icosidodecahedron
5	$\varphi \approx 1.618$	12	0	0	(unbonded)
6	$\sqrt{3} \approx 1.732$	20	0	0	(unbonded)
7	≈ 1.902	12	0	0	(unbonded)
8	2.000	1	0	—	(antipodal)

Theorem 3.1 (Bonded shells of the 600-cell). *The 600-cell has exactly three distance shells with nonzero lattice edges: Shells 1, 2, and 4. Combined with the tetrahedral cells ($4V$, $6E$), there are exactly four polyhedral cage types.*

Proof. By explicit computation of all $\binom{120}{2} = 7140$ pairwise distances and adjacency checks within each shell. $\square$ $\square$

4 The Structural Role of Shell 3

Shell 3 has *zero* lattice edges—no cage can form there. This gap separates the dodecahedron (Shell 2, bottom quark) from the icosidodecahedron (Shell 4, top quark), forcing the top quark’s confinement chains to tunnel through a bondless region.

5 The Cage–Quark Correspondence

Table 2: Cage–quark correspondence.

Cage	600-cell structure	V	E	z	d	Quark
Tetrahedron	Cell	4	6	3	—	Strange
Icosahedron	Shell 1	12	30	5	$1/\varphi$	Charm
Dodecahedron	Shell 2	20	30	3	1.0	Bottom
Icosidodecahedron	Shell 4	30	60	4	$\sqrt{2}$	Top

This correspondence is the unique monotonic bijection from cage vertex count to quark mass.

6 Quark Mass Formulas

6.1 Calibrated formula (v3.x)

Calibrating α from the strange–charm pair:

$$\alpha = \frac{\log(m_c/m_s)}{\log(V_c/V_s)} = \frac{\log(1270/93.4)}{\log(12/4)} = 2.38. \quad (1)$$

With the post-gap multiplier $z = 12$ for the top quark:

Table 3: Calibrated mass predictions (v3.x formula, 2 free params).

Quark	V	Predicted (MeV)	PDG (MeV)	Δ
Strange	4	93 (cal.)	93.4	—
Charm	12	1270 (cal.)	1270	—
Bottom	20	4304	4180	3.0%
Top	30	172 800	172 760	0.02%

6.2 Zero-free-parameter formula (v4.0)

The angular-weighted pair model (SM-9 v2.0) provides three independent derivations that eliminate all free parameters:

1. **The exponent** $\alpha = 7/3$ from the decomposition $V^{7/3} = V^2 \times V^{1/3}$: pair counting ($C(V, 2) \sim V^2$) times linear cage dimension ($V^{1/3}$).
2. **The prefactor** $M_0 = m_e z / \varphi$: the electron mass times the coordination number weighted by the 600-cell edge-to-circumradius ratio $1/\varphi$.
3. **The gap multiplier** $z \times C_F = 16$: Shell 3 forces chains to tunnel via $z = 12$ coordination bonds, each carrying the $SU(3)$ vertex factor $C_F = 4/3$.

Theorem 6.1 (Zero-Parameter Quark Mass Formula).

$$M_q = m_e \frac{z}{\varphi} V_q^{7/3} \times \begin{cases} 1 & q = s, c, b \quad (\text{pre-gap}) \\ z \cdot C_F = 16 & q = t \quad (\text{post-gap}) \end{cases} \quad (2)$$

where $m_e = 0.511 \text{ MeV}$, $z = 12$, $\varphi = (1 + \sqrt{5})/2$, $C_F = 4/3$, and $V \in \{4, 12, 20, 30\}$.

Table 4: Zero-parameter mass predictions (v4.0).

Quark	V	Predicted (MeV)	PDG (MeV)	Δ (%)
Strange	4	96.3	93.4	+3.1
Charm	12	1249	1270	−1.6
Bottom	20	4115	4180	−1.6
Top	30	169 571	172 760	−1.8

RMS error: 2.1% across four orders of magnitude.

6.3 Comparison of v3.x and v4.0

Table 5: Comparison of calibrated (v3.x) and zero-parameter (v4.0) formulas.

Formula	Free params	α	Gap	$m_t \Delta$	RMS
v3.x calibrated	2	2.38 (fitted)	$z = 12$	0.02%	~1%
v4.0 derived	0	7/3	$z \times C_F = 16$	1.8%	2.1%

Remark 6.2 (Reconciliation). *The v3.x and v4.0 formulas are the same physics with different bookkeeping. The exponents differ by only 0.02%. When v3.x calibrates to strange and charm, its fitted M_0 inflates by 1.4% to compensate for the “missing” $C_F = 4/3$ in the gap. The v3.x top quark precision (0.02%) results from a fortunate cancellation. See SM-9 v2.0 for the full analysis.*

6.4 Post-gap coordination multiplier

Theorem 6.3 (Post-Gap Coordination Multiplier — Corrected). *When a cage lies beyond the Shell 3 gap, the self-energy receives a multiplicative factor of $z \times C_F = 12 \times 4/3 = 16$:*

$$m_t = m_e \frac{z}{\varphi} V_t^{7/3} \times z \times C_F = 3.790 \times 30^{7/3} \times 16 = 169\,571 \text{ MeV}. \quad (3)$$

Physical mechanism. For pre-gap quarks, chains follow internal cage edges protected by cage symmetry; the colour algebra is implicitly encoded in $V^{7/3}$. For the top quark, chains must tunnel via the $z = 12$ coordination bonds of the ambient lattice. These external bonds each mediate colour exchange through an hDP propagator carrying the bare $SU(3)$ vertex factor $C_F = 4/3$.

Remark 6.4 (v4.0 correction note). *Version 3.x used $z = 12$ and predicted $m_t = 172,800 \text{ MeV}$ (0.02% error, 2 calibrations). Version 4.0 uses $z \times C_F = 16$ and predicts $m_t = 169,571 \text{ MeV}$ (1.8% error, 0 free params). The trade-off is: 0.02% precision (one parameter) exchanged for 1.8% precision (zero parameters).*

7 The Symmetry Degeneracy Theorem

The angular-weighted pair model (SM-9 v2.0) investigated whether the *angles* between chain-chain pairs within each cage provide additional mass information beyond the vertex count.

Theorem 7.1 (Symmetry Degeneracy). *For vertex-transitive polyhedra on S^2 ,*

$$\sum_{i < j} \sin^2(\theta_{ij}/2) = \frac{V^2}{4}$$

exactly, where θ_{ij} is the angle subtended at the centre by vertices i and j .

This was verified for all four cages. The consequence: angular weighting carries *no information beyond V* for vertex-transitive polyhedra. The physically relevant distinction is the **edge structure**—the bonded fraction $E/C(V, 2)$ drops from 100% (tetrahedron) to 14% (icosidodecahedron).

8 Charge Structure and the 2/3 Ratio

Under balanced charge assignments ($V/2$ positive, $V/2$ negative), the attractive fraction characterises each cage’s oscillatory capacity:

Table 6: Attractive-fraction census.

Cage	V	E	Max	$N_{2/3}$	$P_{2/3}$
Tetrahedron	4	6	2/3	6	100%
Icosahedron	12	30	2/3	72	7.8%
Dodecahedron	20	30	4/5	16 332	8.8%
Icosidodecahedron	30	60	2/3	$\sim 0.7\%$	$\sim 0.7\%$

The ratio $2/3$ is achievable on all four cages: forced on the tetrahedron, maximal on the icosahedron and icosidodecahedron. Whether this $2/3$ and the Koide ratio $K = 2/3$ (SM-3) share a geometric origin is an open question.

9 Palindrome Structure and Three Generations

The distance shells exhibit mirror symmetry: Shell 7 mirrors Shell 1, Shell 6 mirrors Shell 2, Shells 3 and 5 are both gaps, with Shell 4 at the midpoint.

Theorem 9.1 (Three generations from lattice tessellation). *The tessellated 600-cell lattice supports exactly four independent cage types, corresponding to exactly three generations of quarks. No fourth generation exists because outer shells are identified with inner shells of neighbouring 600-cells.*

Proof. Shell 7 of centre A is Shell 1 of a neighbouring centre B . Any cage at that location is an icosahedral cage of B , not a new species. The four independent cage types map to up/down (no cage), strange/charm (tet/ico), bottom/top (dod/icosidod). Shells 5–7 generate no new species. $\square$ $\square$

This prediction is falsifiable: a fourth-generation quark would refute the antipodal identification.

10 Anticipated Criticisms

“Is the exponent $7/3$ derived or fitted?” Partially derived: $V^{7/3} = V^2 \times V^{1/3}$ (pair count $\times$ cage dimension). A rigorous proof from simplex combinatorics remains open.

“Why does m_e appear?” The electron mass sets the DP Sea energy scale. All quark masses derive from ZBW oscillation energy anchored to m_e through SM-6.

“Is the $z \times C_F$ multiplier ad hoc?” No. Both $z = 12$ (lattice geometry) and $C_F = 4/3$ (SU(3) algebra) are independently established. The multiplier activates at a definite geometric feature (Shell 3 gap).

“Why not keep $z = 12$ with 0.02% precision?” Because v3.x’s 0.02% precision requires two calibrations. The v4.0 formula achieves 2.1% with zero parameters. The v3.x precision is partly a cancellation artefact (SM-9 v2.0).

11 Open Problems

1. Rigorous proof of $\alpha = 7/3$ from cage geometry.
2. Derive the EW feedback correction $\varepsilon \approx \alpha_{\text{geom}}/z^2$ in the exponent (SM-9 v2.0).
3. Strange quark residual (+3.1%).
4. Unified cage + Koide framework.
5. Derive the attractive-fraction selection principle.

12 Conclusion

The 600-cell contains exactly four bonded shells, separated by a structurally significant gap at Shell 3. This paper establishes two mass formulas from this geometry: a calibrated formula ($V^{2.38 \times z}$, 0.02% top precision, 2 parameters) and a zero-free-parameter formula ($m_e(z/\varphi)V^{7/3} \times zC_F$, 2.1% RMS, 0 parameters). Both are the same physics with different bookkeeping.

The zero-parameter formula is the first in CPP where 600-cell geometry (z, φ, V), SU(3) colour algebra (C_F), and the electron mass (m_e) appear together in a single mass prediction. The palindrome structure and antipodal identification limit the Standard Model to exactly three generations.

Remark 12.1 (Epistemological status). *SM-8 v4.0 promotes the quark mass hierarchy from calibrated coherence (v3.x: exact geometry + empirical scaling) to derived coherence (v4.0: exact geometry + partially derived scaling + zero free parameters). The structural results (shell embedding, palindrome, generation count, charge census) are exact and independent of the mass formula.*

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Thomas Lee Abshier, ND conceived the cage hierarchy model, proposed the electron capture mechanism for down-type quarks, identified the impedance boundary picture, recognised the Shell 3 gap significance, directed the pair-model investigation, and provided the physical vision throughout.

Claude Opus (Anthropic) computed the 600-cell distance shells, identified the four bonded shells, performed the charge census, developed the angular-weighted pair model, proved the symmetry degeneracy theorem, derived the zero-parameter formula with $M_0 = m_e z / \varphi$ and $\text{gap} = z \times C_F = 16$, and drafted all versions.

Grok (xAI) provided the original post-gap multiplier $z = 12$, conjectured $\alpha = 3 - 1/\varphi$, and co-developed the cage hierarchy.

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The CPP programme is registered at OSF (DOI: <https://doi.org/10.17605/OSF.IO/JXE8D>) and maintained at GitHub (<https://github.com/Hyperphysics-Institute/Cpp>).

A Physical Mechanism of the Post-Gap Multiplier

The post-gap coordination multiplier (Theorem 6.3) has a precise physical mechanism rooted in the near-field/far-field transition of ZBW chain oscillations.

A.1 Near-field regime (Shells 1–2)

For cages at Shells 1 and 2, the cage surface is close to the central CP. A ZBW signal launched from the central CP reaches the cage surface and reflects back within one oscillation period. The reflected signal (back-EMF) arrives in phase with the source, establishing a stable standing-wave resonance. This back-EMF cancels the driving signal, preventing additional chain formation. The equilibrium state supports a limited number of radial chains (typically 4, arranged tetrahedrally to minimise inter-chain repulsion).

A.2 Far-field regime (Shell 4, beyond the gap)

For the top quark cage at Shell 4 ($d = \sqrt{2} \approx 1.414$), the cage surface is separated from the central CP by the Shell 3 gap. Shell 3 contains 12 Grid Points but zero lattice edges — no bonded structure exists there. A ZBW signal reaching Shell 3 encounters isolated Grid Points with no coherent reflecting surface. The signal passes through Shell 3 and continues to Shell 4, where it finally reflects off the bonded icosidodecahedral cage.

The round-trip distance ($2 \times 1.414 = 2.83$ lattice units) is sufficiently large that the reflected signal arrives out of phase with the source oscillation. The back-EMF is delayed beyond the coherence time. The suppression mechanism that normally limits chain formation to 4 channels breaks.

A.3 Channel saturation

Without prompt back-EMF, the central CP is free each Moment to initiate ZBW attraction along whichever of the $z = 12$ lattice bonds is locally most favourable. Over time, all 12 channels accumulate ZBW impulses in various stages of transit between the central CP and the Shell 4 cage surface. The internal volume fills with asynchronous oscillations across all 12 directions.

The mass multiplier engages all $z = 12$ channels because: (a) the 600-cell is 12-regular, providing exactly 12 channels from every vertex; (b) the far-field condition lifts the back-EMF suppression uniformly across all channels; and (c) the Shell 3 gap ensures no intermediate reflecting surface restores the near-field condition.

Remark A.1 (v4.0 correction). *The v3.x analysis attributed the full multiplier to the coordination number alone ($z = 12$). The v4.0 angular-weighted pair model shows that each of the 12 coordination bonds also carries the $SU(3)$ colour vertex factor $C_F = 4/3$, giving a total multiplier of $z \times C_F = 16$. The near-field/far-field mechanism (which explains why all 12 channels activate) remains valid; the Casimir factor (which explains the weight per channel) is an additional algebraic contribution identified in the pair model analysis.*

B The Impedance Boundary at Cage Vertices

A displacement wave propagating along a qDP chain (gluon bond) within a cage encounters each cage vertex as an impedance boundary. The vertex CP has mass (inertia), causing partial reflection and partial transmission of the wave.

The **reflection fraction** determines confinement: heavier quarks reflect more colour energy back into the cage, producing stronger confinement. The **transmission fraction** determines the nuclear force: energy that passes through a quark vertex radiates into the DP Sea as SSV perturbations, which neighbouring baryons feel as the residual strong force.

This impedance picture provides a mass-dependent confinement mechanism without requiring a separate confinement axiom. The top quark's Shell 4 cage, with its large radius and far-field dynamics, has a qualitatively different impedance structure from the smaller cages, contributing to its failure to hadronize in the usual manner.

C Charge Structure and the Repulsive-Edge Reflector

The hybrid tetrahedron (the baryon scaffold) has 4 attractive ZBW edges and 2 repulsive edges. Every triangular face has exactly 1 repulsive edge and 2 attractive edges. The repulsive edges act as reflective boundaries for displacement waves, trapping energy on each face and driving face-mode circulation.

This dynamic mechanism supplements the kinematic Walk-Dimension argument: face modes circulate not merely because length-3 walks exist, but because the charge structure of the cage creates reflective walls that physically confine displacement pulses to circulating patterns. The 2 attractive edges are waveguides; the 1 repulsive edge is a mirror. Each face is a natural oscillation cavity.

References

Thomas Lee Abshier, Grok (xAI), Claude Opus (Anthropic), and Copilot (Microsoft). The Charged Lepton Mass Spectrum from 600-Cell Lattice Geometry. *SM-6 v3, Hyperphysics Institute*, 2026a.

Thomas Lee Abshier, Grok (xAI), Claude Opus (Anthropic), and Copilot (Microsoft). Heavy Quark Mass Spectrum and Strong Coupling from 600-Cell Lattice Geometry. *SM-7 v2.2, Hyperphysics Institute*, 2026b.